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Genome-wide identification and expression analysis of MAPK and MAPKK gene family in *Malus domestica*

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ABSTRACT

MAPK signal transduction modules play crucial roles in regulating many biological processes in plants, which are composed of three classes of hierarchically organized protein kinases, namely MAPKKKs, MAPKKs, and MAPKs. Although genome-wide analysis of this family has been carried out in some species, little is known about *MAPK* and *MAPKK* genes in apple (*Malus domestica*). In this study, a total of 26 putative apple *MAPK* genes (*MdMPKs*) and 9 putative apple *MAPKK* genes (*MdMKKs*) have been identified and located within the apple genome. Phylogenetic analysis revealed that *MdMAPKs* and *MdMAPKKs* could be divided into 4 subfamilies (groups A, B, C and D), respectively. The predicted *MdMAPKs* and *MdMAPKKs* were distributed across 13 out of 17 chromosomes with different densities. In addition, analysis of exon–intron junctions and of intron phase inside the predicted coding region of each candidate gene has revealed high levels of conservation within and between phylogenetic groups. According to the microarray and expressed sequence tag (EST) analysis, the different expression patterns indicate that they may play different roles during fruit development and rootstock–scion interaction process. Moreover, *MAPK* and *MAPKK* genes were performed expression profile analyses in different tissues (root, stem, leaf, flower and fruit), and all of the selected genes were expressed in at least one of the tissues tested, indicating that the *MAPKs* and *MAPKKs* are involved in various aspects of physiological and developmental processes of apple. To our knowledge, this is the first report of a genome-wide analysis of the apple *MAPK* and *MAPKK* gene family. This study provides valuable information for understanding the classification and putative functions of the MAPK signal in apple.

1. Introduction

As sessile organisms, plants have developed complex signaling networks to sense environmental signals and adapt themselves to environmental stresses. Plants transmit the extracellular signal to intracellular reaction through a typical mechanism. A common mechanism to translate such external stimuli into cellular responses is the activation of mitogen-activated protein kinase (MAPK) cascades, which consist of three subsequently protein kinases: the MAP kinase kinase kinase (MAPKKK, MKKK, MEKK, MAP3K), MAP kinase kinase (MAPKK, MKK, MEK, MAP2K) and MAP kinase (MAPK, MPK) (MAPK Group, 2002). MAPKKKs are serine/threonine kinases phosphorylating two amino acids in the S/T-X_{3-5}-S/T motif of the MKK activation loop. MKKs are dual-specificity kinases that activate a MAPK through double phosphorylation of the T-X-Y motif in the activation loop (MAPK Group, 2002). MAPKs are serine/threonine kinases that phosphorylate various cytoplasmic or nuclear substrates, including other kinases and transcription factors (Bethke et al., 2009; Fiil et al., 2008; Meldau et al., 2012; Popescu et al., 2009; Sasabe et al., 2006; Shen et al., 2012; Xu et al., 2008; Yoo et al., 2008). MAPK substrates play crucial roles in responding stimulation and regulating a variety of important intracellular reactions.

MAPK cascades are universal signal transduction modules in eukaryotes, including animals, yeasts, and plants. When exposed to a various abiotic and biotic stresses, such as drought, low temperature, high salt, mechanical damage, osmotic stress, oxidative stress and pathogen infection, the MAPK cascade is rapidly activated (Burnett et al., 2000; Cardinale et al., 2002; Colcombet and Hirt, 2008; Desikan et al., 2001; Ichimura et al., 2000; Kiegerl et al., 2000; Kumar et al., 2008; Zhang et al., 2012a; Liu, 2012; Mockaitis and Howell, 2000; Munnik and Meijer, 2001; Raina et al., 2012; Ren et al., 2002; Rodriguez et al., 2010; Teige

Abbreviations: MAPK, mitogen-activated protein kinase; MAPKK, MAPK kinase; BLAST, Basic Local Alignment Search Tool; GWD, Genome-Wide Duplication; HMM, Hidden Markov Model; MEGA, Molecular Evolutionary Genetics Analysis; MUSCLE, Multiple Sequence Comparison by Log-Expectation; NJ, neighbor-joining; SMART, Simple Modular Architecture Research Tool; Apple GFDB database, Apple Gene Function and Gene Family Database; GDR, Genome Database for Rosaceae; Pfam, Protein family; GSDS, Gene Structure Display Server; NCBI, National Center for Biotechnology Information; ORF, open reading frame; MW, molecular weight; aa, amino acid; pIs, isoelectric points.

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et al., 2004; Yuasa et al., 2001; Z. Zhang et al., 2012). Based on the phylogenetic analysis of amino acid sequence and phosphorylation motif, plant MAPKs can be divided into four subfamilies (A, B, C, D). Members of A, B, and C subfamilies have a TEY phosphorylation motif in their active sites, while members in D subfamily have a TDY motif in their active sites and have a long C terminal sequence, which has higher homology with mammals and yeast p38, such as AtMPK8, AtMPK9 and BWMK1 (MAPK Group, 2002). At present, the most extensively studied member of group D MAPKs in plants is OsBWMK1 (Cheong et al., 2003; Koo et al., 2007). The amino acid sequence of MKKs phosphorylation sites in plants is different from that of mammals. Plant MKKs contain conserved S/TxxxxxS/T activation motif, whereas mammalian enzymes have S/TxxxS/T motif. However, the conserved sequence in AtMKK10 is missing (Hamel et al., 2006). The N-terminal of MKKs in plants has a MAPK docking site, namely (K/R)-K/R-K/R-x (2, 7)-L/I/V-x-L/I/V-x-(L/I/V) (MAPK Group, 2002). However, the canonical motif is missing in AtMKK3, ZmMAPKK1 and AtMKK8 (Kumar et al., 2008; MAPK Group, 2002).

In plant, the most extensively studied MAPKs are *AtMPK3/4/6* in *Arabidopsis* and their orthologs in other plant species. MKK2–MPK4/MPK6 cascade was shown to be activated by cold and salt stresses (Teige et al., 2004). MEKK1–MKK1/MKK2–MPK4–MKS1/WRKY33 cascade negatively controls plant immune response (Berriri et al., 2012; Gao et al., 2008). More recently, it is also reported that MEKK1–MKK1/MKK2–MPK4 cascade negatively regulates SUMM1 (MEKK2) which functions as a positive regulator of SUMM2-mediated immune response (Kong et al., 2012a; Zhang et al., 2012a; Kong et al., 2012b; Zhang et al., 2012b). AtMKK1–AtMPK6 mediates ABA-induced *CAT1* expression and H_2O_2 production (Xing et al., 2008). MEKK1–MKK4/5–MPK3/6–WRKY 22/WRKY29 plays an important role in plant innate immunity (Asai et al., 2002). MKK9–MPK3/MPK6 regulates ethylene signaling and camalexin biosynthesis and may also play a role in leaf senescence (Ren et al., 2008; Xu et al., 2008; Yoo et al., 2008). ANP2/3–MKK6–MPK4/11/13 plays roles in the regulation of cytokinesis (Beck et al., 2011; Takahashi et al., 2010). In tobacco, overproduction of a constitutively active form of tobacco NtMEK2 caused hypersensitive cell death through activation of SIPK and WIPK (Ren et al., 2002). In rice, OsMKK4–OsMPK3 is involved in arsenic stress signal transduction (Rao et al., 2011). A chitin elicitor is shown to activate OsMKK4–OsMPK3/OsMPK6 and it is essential for the biosynthesis of diterpenoid phytoalexins induced by chitin elicitors (Kishi-Kaboshi et al., 2010).

To date, the members of MAPK cascades from *Arabidopsis* and rice have been identified. Analysis of *Arabidopsis* genome revealed the presence of 20 MAPKs, 10 MKKs, and around 80 MAPKKKs, and in silico analysis of rice genome revealed 17 MAPKs, 8 MKK and 75 MAPKKKs (Hadiarto et al., 2006; Hamel et al., 2006; Kumar et al., 2008; MAPK Group, 2002; Rao et al., 2010; Singh et al., 2012). More recently, a total of 16 MAPK genes and 12 MKK genes were identified from *Brachypodium distachyon* (Chen et al., 2012). Apple (*Malus* × *domestica*) is one of the most widely cultivated fruit trees, and is the most economically important woody plant in temperate regions (Lee et al., 2007). In contrast to the intensive research on MAPK and MAPKK in both model and crop plants such as *Arabidopsis*, rice and poplar, there are only very limited reports on the characterization of MAPK and MAPKK in apple. Recently, the draft genome sequence of apple has been decoded, which provided an excellent opportunity for genome-wide analyses of all the genes belonging to specific gene families (Velasco et al., 2010). The genome-wide analysis of the *RING finger* gene family, *DREB* gene family, *dehydrin* gene family and *Hsf* gene family have been reported in apple (Giorno et al., 2012; Li et al., 2011; Liang et al., 2012; Zhao et al., 2012). However, no genome-wide information on the apple MAPK and MAPKK gene family is currently available.

Given the importance of MAPK and MAPKK in diverse biological and physiological processes and their potential application for the development of improved stress-tolerant transgenic plants, we carried out a systematic analysis of the apple MAPK and MAPKK family in the present study for the first time. And then the chromosome location, gene structure of the putative MAPK and MAPKK genes predicted by genome-wide surveys of the apple genomic sequences were carefully analyzed. Additionally, the putative MAPK and MAPKK were subjected to phylogenetic analyses with their *Arabidopsis* counterparts. These comparisons have enabled the identification of gene orthologs and clusters of orthologous groups that can be studied for further functional characterization. Furthermore, we analyze the expression patterns using microarray and expressed sequence tag (EST) data. Semi-quantitative RT-PCR was performed to reveal the expression profile in different tissues (root, stem, leaf, flower and fruit). To our knowledge, this is the first reported genome-wide analysis of the apple MAPK and MAPKK family, which would provide valuable information for understanding the classification and putative functions of MdMAPKs and MdMAPKKs. Ultimately, these findings will lead to potential applications for the improvement of stress resistance in apple via genetic engineering.

2. Materials and methods

2.1. Identification of MdMAPKs and MdMAPKKs in apple

To identify members of the *MAPK* and *MAPKK* gene family, multiple database searches were performed. The *Arabidopsis* and rice *MAPK* gene sequences were used as queries to perform repetitive blast searches against the GDR database (Genome Database for Rosaceae: http://www.rosaceae.org/). Additionally, all protein sequences were then used as queries to perform multiple database searches against proteome and genome files downloaded from GDR database. Stand-alone versions of BLASTP and TBLASTN (Basic Local Alignment Search Tool: http://blast.ncbi.nlm.nih.gov) available from NCBI (National Center for Biotechnology Information) were used with the e-value cutoff set to 1e-003 (Altschul et al., 1990). Moreover, the predicted *MAPK* and *MAPKK* gene family sequences were downloaded from the Apple GFDB database (Apple Gene Function and Gene Family Database: http://www.applegene.org/). All protein sequences derived from the candidate MAPK genes collected were examined using the domain analysis programs, Pfam (Protein family: http://pfam.sanger.ac.uk/) and SMART (Simple Modular Architecture Research Tool: http://smart.embl-heidelberg.de/) with the default cut off parameters (Bateman et al., 2002; Letunic et al., 2012). The isoelectric points and molecular weights of *MdMAPKs* and *MdMAPKKs* were obtained with the help of proteomics and sequence analysis tools on the ExPASy Proteomics Server (http://expasy.org/).

2.2. The chromosomal location and gene structure of MdMAPK and MdMAPKK genes

The chromosomal locations and gene structures were retrieved from the apple genome data that were downloaded from the GDR database. The remaining genes were selected using a Perl-based program and mapped to the chromosomes with MapDraw, as well as the gene structures of the *MdMAPK* and *MdMAPKK* were generated with the GSDS (Gene Structure Display Server: http://gsds.cbi.pku.edu.cn/).

2.3. Sequence alignment and phylogenetic analysis

MdMAPK and *MdMAPKK* sequences were aligned using the program ClustalX with BLOSUM30 as the protein weight matrix. The MUSCLE (Multiple Sequence Comparison by Log-Expectation) program (version 3.52) was also used to perform multiple sequence alignments to confirm the ClustalX data output (http://www.clustal.org/) (Edgar, 2004). Phylogenetic trees based on the protein sequences of the *MdMAPK* and *MdMAPKK* were constructed using the NJ (neighbor-joining) method of the program MEGA5 (Molecular Evolutionary Genetics Analysis) with p-distance and the complete deletion option parameters engaged (Saitou and Nei, 1987; Tamura et al., 2011). The reliability of the trees

Table 1
The information of *MAPK* and *MAPKK* gene family in apple.

Gene family	Gene	Gene model	Homologous gene	Genomic position	ORF (bp)	Size (aa)	MW (Da)	pIs
MAPK	MdMPK1-1	MDP0000128473	ATMPK1	chr8: 1626465…1628251	1119	372	42545	6.08
	MdMPK1-2	MDP0000165532	ATMPK1	chr8: 1631845…1633632	1119	372	42545	6.08
	MdMPK1-3	MDP0000210110	ATMPK1	chr12: 21732164…21733860	1119	372	42587	6.56
	MdMPK3-1	MDP0000199036	ATMPK3	chr11: 10355188…10357200	1113	370	42773	5.62
	MdMPK3-2	MDP0000237742	ATMPK3	chr3: 9421561…9423600	1113	370	42619	5.83
	MdMPK3-3	MDP0000321850	ATMPK3	chr11: 10355015…10357027	1113	370	42773	5.62
	MdMPK4-1	MDP0000251955	ATMPK4	chr1: 9998861…10002694	1122	373	42977	6.14
	MdMPK4-2	MDP0000321746	ATMPK4	chr1: 9996933…10006251	2397	798	90222	6.44
	MdMPK4-3	MDP0000326020	ATMPK4	chr1: 10004130…10008154	1125	374	42981	6.22
	MdMPK4-4	MDP0000766240	ATMPK4	Not found	1134	377	43084	6.17
	MdMPK6-1	MDP0000321308	ATMPK6	chr2: 1258618…1261990	1212	403	46253	5.83
	MdMPK6-2	MDP0000340624	ATMPK6	chr15: 7772120…7775308	1224	407	46385	5.66
	MdMPK7-1	MDP0000807889	ATMPK7	chr2: 12718995…12720478	1137	378	43504	7.65
	MdMPK7-2	MDP0000826016	ATMPK7	chr2: 12521296…12522781	1137	378	43614	8.04
	MdMPK9	MDP0000294142	ATMPK9	chr14: 28270635…28274646	1908	635	72254	7.71
	MdMPK13-1	MDP0000422421	ATMPK13	chr11: 10226795…10231896	1458	485	55078	5.38
	MdMPK13-2	MDP0000593502	ATMPK13	chr3: 9282219…9284697	1125	374	42902	5.07
	MdMPK16-1	MDP0000170804	ATMPK16	chr13: 17406449…17411267	2028	675	76111	7.05
	MdMPK16-2	MDP0000879089	ATMPK16	chr16: 13839761…13843295	1767	588	66718	8.26
	MdMPK17	MDP0000277562	ATMPK17	chr13: 1536570…1539982	1752	583	66483	6.51
	MdMPK19-1	MDP0000121116	ATMPK19	chr3: 29166863…29171294	1827	608	69032	9.1
	MdMPK19-2	MDP0000169216	ATMPK19	chr3: 29118838…29123334	1893	630	71642	8.76
	MdMPK19-3	MDP0000195781	ATMPK19	chr3: 29241439…29245728	1851	616	69919	9.33
	MdMPK19-4	MDP0000233021	ATMPK19	chr3: 29196021…29200484	1884	627	71812	9.21
	MdMPK20-1	MDP0000188369	ATMPK20	chr17: 12043155…12046693	1827	608	69769	9.54
	MdMPK20-2	MDP0000250639	ATMPK20	chr9: 11741310…11747719	2598	865	98916	9.2
MAPKK	MdMKK2	MDP0000320149	ATMKK2	chr2: 6496133…6500989	966	321	36020.28	6.18
	MdMKK3	MDP0000293246	ATMKK3	chr9: 1608365…1610979	1665	554	61499.15	5.27
	MdMKK4-1	MDP0000185340	ATMKK4	chr9: 12296305…12297375	1071	356	39446.82	9.06
	MdMKK4-2	MDP0000469046	ATMKK4	chr17: 12389463…12390536	1074	357	39387.83	9.2
	MdMKK6-1	MDP0000161427	ATMKK6	chr2: 5460568…5463710	1236	411	46334.23	6.13
	MdMKK6-2	MDP0000332757	ATMKK6	chr15: 14051008…14054220	1086	361	39742.52	5.46
	MdMKK9-1	MDP0000252992	ATMKK9	chr14: 28591845…28592789	945	314	35502.73	7.12
	MdMKK9-2	MDP0000459132	ATMKK9	chr16: 10245405…10246355	951	316	34677.77	6.86
	MdMKK9-3	MDP0000932944	ATMKK9	chr6: 24077483…24078451	969	322	36212.75	7.08

obtained was tested using bootstrapping with 1000 replicates. Images of the phylogenetic trees were also drawn using MEGA5.

2.4. Expression analysis of MdMAPKs and MdMAPKKs in microarray

The microarray data of gene expression in apple fruit during fruit ripening process was downloaded from the Gene Expression Omnibus database using the GSE series accession number GSE24523. The sequences of identified *MdMAPK* and *MdMAPKK*-containing genes were used as queries to blast against probe sequence (GPL11164) to find the corresponding unigene IDs used in microarray data. The microarray data during rootstock–scion interaction process (GSE4762) was downloaded from the Gene Expression Omnibus database, too. And *MdMAPK* and *MdMAPKK*-containing genes were used as query to blast against probe platform (GPL3715) to find the corresponding unigene IDs used in microarray data. Phylogenetic analysis was performed to determine the corresponding unigene IDs when sequences of high similarity were acquired. The microarray data were made into a database by a Perl-based program, then clustered using Cluster3.0 with Euclidean distances and the hierarchical cluster method of complete linkage clustering. The clustering tree was constructed and viewed in Java Treeview.

2.5. Plant materials and gene expression analysis

Root, stem, leaf, flower and fruit samples from apple (*Malus hupehensis* Rehd.) were collected from a 5-year-old apple tree that was grown in natural conditions in the Shandong Province of China and stored at −80 °C. RNA was extracted from triplicate biological replicates of the above samples using the CTAB method followed by treatment with Turbo DNase I (Ambion, Austin, TX), and reverse transcription of 3 μg RNA from each sample. First-strand cDNA was synthesized using the PrimeScript 1st Strand cDNA Synthesis Kit (Takara, Dalian, China). The transcript levels of the *MdMAPK* genes were determined by semi-quantitative RT-PCR. Details of primers are listed in the Supplemental Table S1.

3. Results

3.1. Identification of MAPK and MAPKK genes in apple

To identify MAPK encoding genes from apple genome, BLASTP searches of the entire apple genome database (GDR) using other three well-studied plant (*Arabidopsis*, rice and poplar) MAPKs as queries were firstly performed. The HMM of the SMART and Pfam tools was then exploited as query to confirm the putative MAPK genes. Finally, 26 typical *MAPK* genes containing full ORFs were identified and manually analyzed using InterProScan and ClustalX program to confirm the presence of MAPK domain, and were used for further analysis below. In order to distinguish from the remaining *MAPKs*, we provisionally named them as *MdMPK1-1*, *MdMPK1-2* or *MdMPK1-3*, based on the best homologous gene in *Arabidopsis* (Table 1). The open reading frame (ORF) length ranged from 1113 bp (*MdMPK3-1*) to 2598 bp (*MdMPK20-2*), with an average ORF length of 1401 bp. The identified *MdMAPK* genes encode proteins ranging from 370 (*MdMPK3-1*, *MdMPK3-2* and *MdMPK3-3*) to 865 (*MdMPK20-2*) amino acids (aa) in length with an average of 500 aa, with a predicted molecular mass range of 42, 545–98, and 916 Da, and protein pIs ranging from 5.07 (*MdMPK13-2*) to 9.54 (*MdMPK20-1*) (Table 1).

Using the same methods, nine MAPKK genes were identified from the apple genome, and named as *MdMPKKs*, which encode proteins ranging from 314 (*MdMKK9-1*) to 554 (*MdMKK3*) amino acids (aa) in length with an average of 368 aa, and protein pIs ranging from 5.27 (*MdMKK3*) to 9.2 (*MdMKK4-2*) (Table 1).

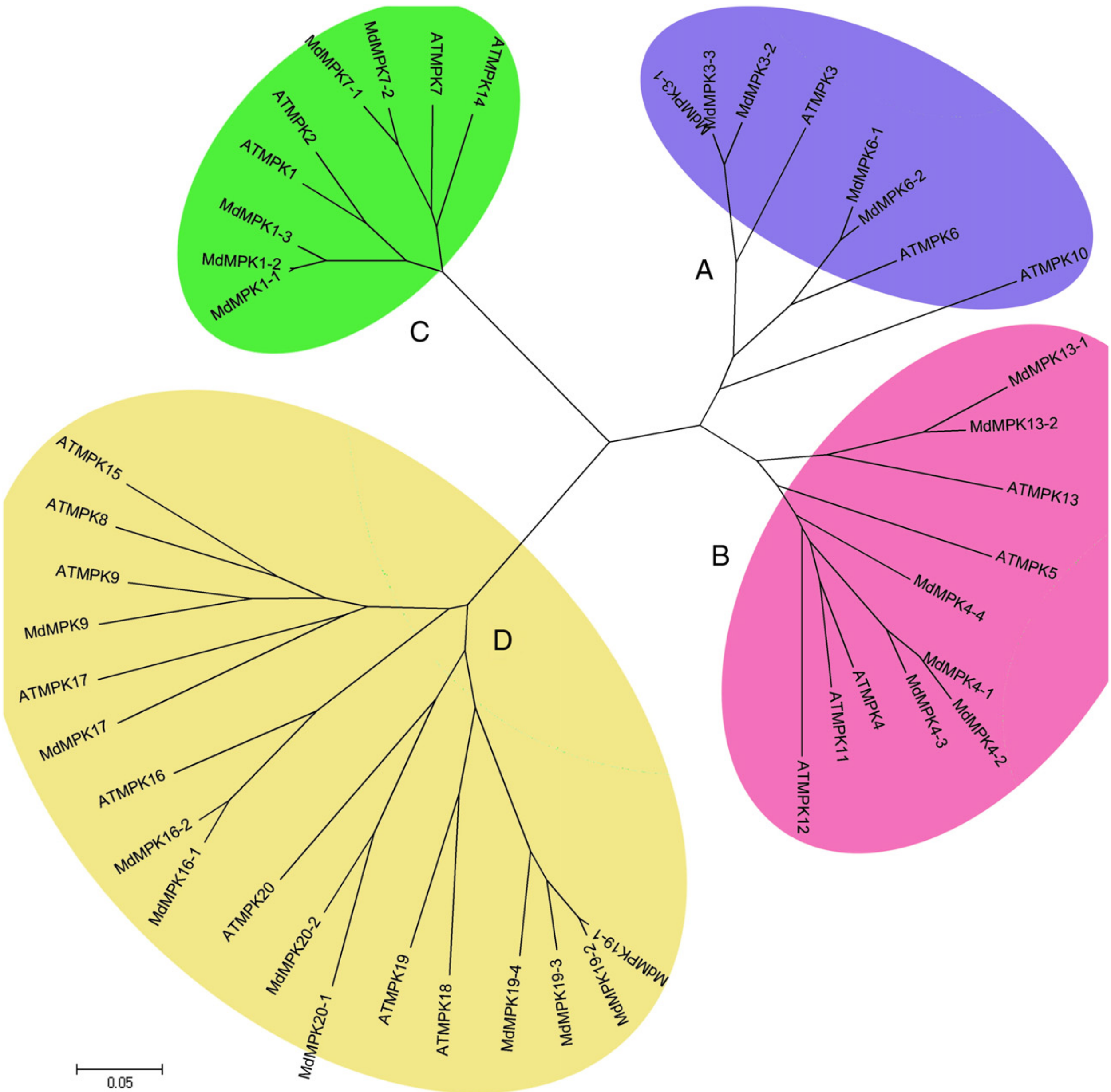

Fig. 1. Phylogenetic relationship of *Arabidopsis* and apple *MAPK* genes. The phylogenetic tree was constructed based on a complete protein sequence alignment of *MAPKs* in *Arabidopsis* and apple by the neighbor-joining method with bootstrapping analysis (1000 replicates). The subgroups are marked by the colorful background. Scale bar represents 0.05 amino acid substitution per site.

Table 2
The number of the *MAPK* and *MAPKK* gene family in *Arabidopsis*, rice, poplar, tomato, *Brachypodium distachyon* and apple.

Gene family	Species	Group A	Group B	Group C	Group D	Total
MAPK	*Arabidopsis*	3	5	4	8	20
	Rice	2	1	2	10	15
	Poplar	4	4	4	9	21
	Tomato	3	4	2	7	16
	B. distachyon	2	2	3	9	16
	Apple	5	6	5	10	26
MAPKK	*Arabidopsis*	3	1	2	4	10
	Rice	2	1	2	3	8
	Poplar	3	1	2	5	11
	B. distachyon	2	3	2	5	12
	Apple	3	1	2	3	9

3.2. Phylogenetic relationships of MAPK and MAPKK gene family in apple

In order to evaluate the evolutionary relationship among the *MdMAPK* proteins, full-length amino acid sequences of 26 MdMAPKs and 20 AtMAPKs from *Arabidopsis* were subjected to a multiple sequence alignment using the MEGA5 program. The multiple sequence alignment file was then used for the construction of an unrooted phylogenetic tree. As shown in Fig. 1, the phylogenetic tree divided the MAPKs into four groups (groups A, B, C and D) as monophyletic clades with at least 50% bootstrap support. Nine sister pairs and a cluster of paralogous *MdMAPK* genes are found, and marked by red shadow, which have very strong bootstrap support (>90%) (Supplemental Fig. 1). Both groups A and C contain 5 apple genes, and group D constitutes the largest clade containing 10 *MdMAPKs*, followed by group B (6 genes). In apple, MAPKs carry either a TEY motif in members of groups A, B, and C or a TDY motif in members of group D in their active

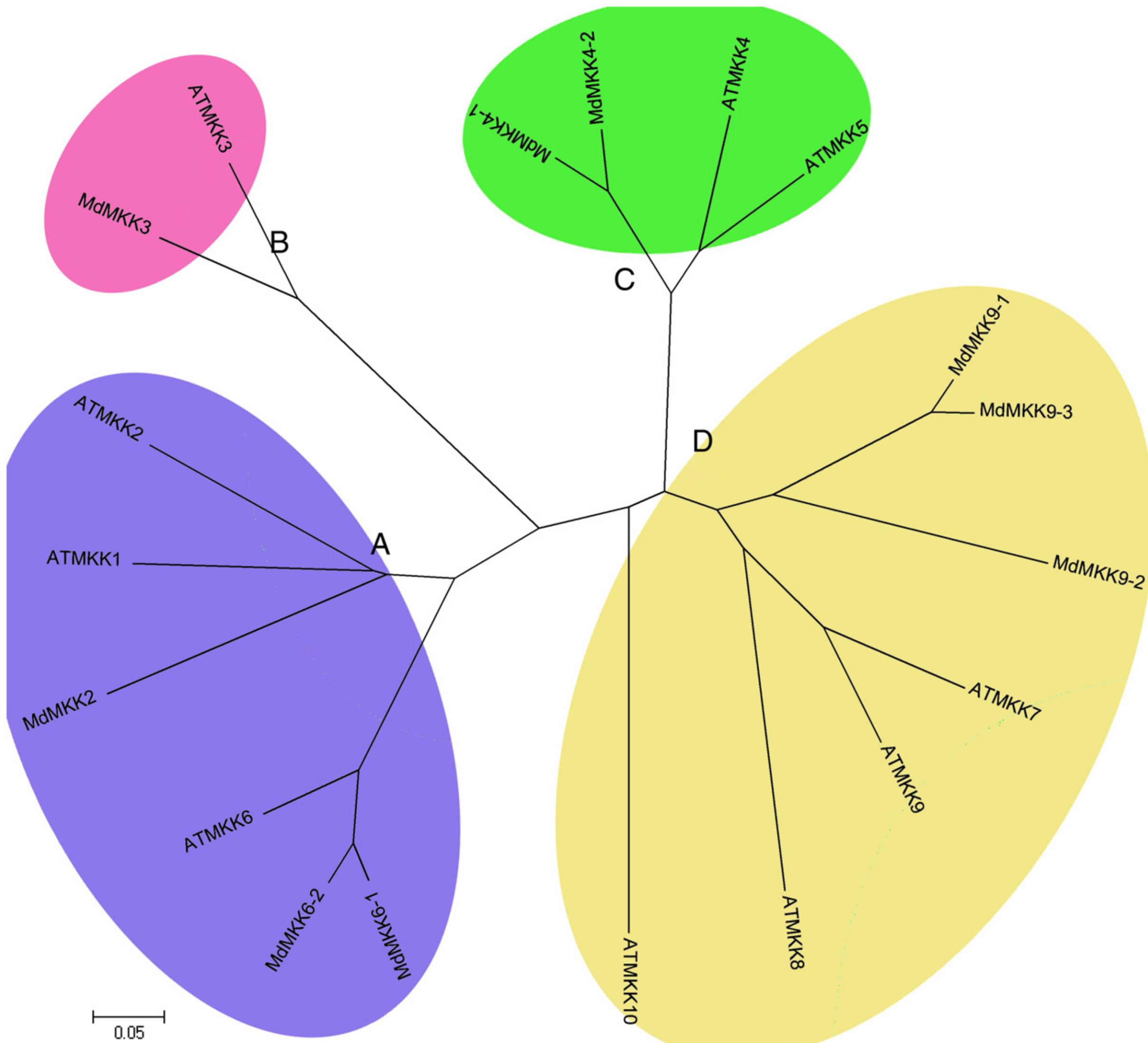

Fig. 2. Phylogenetic relationships of *Arabidopsis* and apple *MAPKK* genes. The phylogenetic tree was constructed based on a complete protein sequence alignment of *MAPKKs* in *Arabidopsis* and apple by the neighbor-joining method with bootstrapping analysis (1000 replicates). The subgroups are marked by the colorful background. Scale bar represents 0.05 amino acid substitution per site.

sites (Supplemental Fig. 2). As shown in Table 2, the group D gene family is the largest subfamily in *Arabidopsis*, rice, poplar and tomato, too. The size of groups A and B is similar in *Arabidopsis*, poplar, tomato and apple, concluding more genes than in rice, which shows that the amplification of groups A and B gene numbers is particularly impressive in the four dicotyledons (*Arabidopsis*, poplar, tomato and apple). The main reason underlying this difference is the existence of a large number of tandem duplications in dicotyledons genome.

Only nine *MAPKK* genes were identified from the apple genome, and divided into four subfamilies (groups A, B, C and D) by the phylogenetic analysis. Three *MdMAPKK* sister pairs were found by strong bootstrap support (>90%), and marked by red shadow (Supplemental Fig. 3). Three, one, two and three genes fall into the groups A, B, C and D, respectively. In comparison with other plants (*Arabidopsis*, rice and poplar), it is found that the number of *MAPKK* gene family is very conserved (Table 2).

3.3. Exon and intron organization of MdMAPK and MdMAPKK genes

Structural analyses are supposed to provide valuable information concerning duplication events when interpreting phylogenetic relationships within gene families. In *MdMAPK* gene family, the number of introns varied from 2 (five of group C *MdMAPKs*) to 15 (*MdMPK20-2*, a group D MdMAPK gene) (Fig. 3). Additionally, most members within the same subfamily shared a similar exon/intron structure and gene length. In group A, three *MdMPKs* consist of six exons, and two genes have five exons with the longer intron. According to genome-wide analysis of other plants, we found that all group A *MAPKs* consisted 6 exons in *Arabidopsis*, poplar and tomato. *MdMPKs* belonging to group B possess six exons, with the exception of 5 exons in *MdMPK4-2* and *MdMPK13-1*. In tomato and poplar, group B genes contain six exons, while four exons in three MAPKs (AtMPK5, AtMPK11 and AtMPK13) in *Arabidopsis*. Group C *MdMAPKs* are composed of only two exons with strictly conserved sizes each, which are very similar to the other plant, including *Arabidopsis*, poplar and tomato. In contrast to the highly conserved structural patterns in group C, group D *MdMPKs* possess a complex distribution of exons and introns, including different pattern subsets within the same phylogenetic group. For instance, *MdMPK20-2* possesses 15 exons, *MdMPK16-1* possesses 12 exons, *MdMPK17* and *MdMPK9* are both composed of 11 exons, while the rest of the members posses ten (Fig. 3). Likewise, the group D MAPKs is complex in *Arabidopsis*, tomato and poplar. Despite of some modest differences in the length of particular exons, it is clear that the exon structural pattern

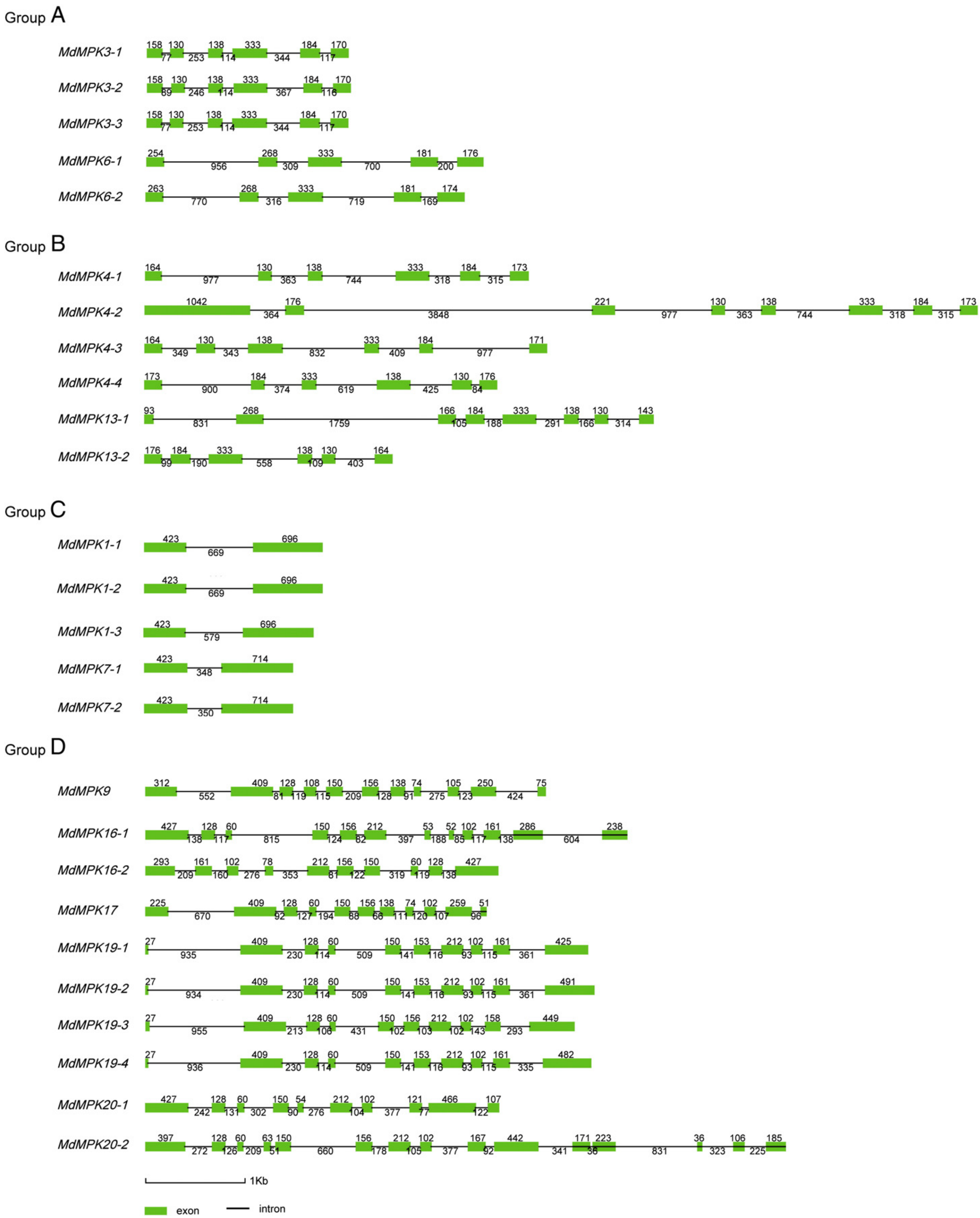

Fig. 3. The exon/intron structure of Apple *MAPK* genes. Introns and exons are represented by black lines and green boxes respectively. The length in base pairs of each intron and exon is also indicated. Numbers correspond to the length of the intron and exon.

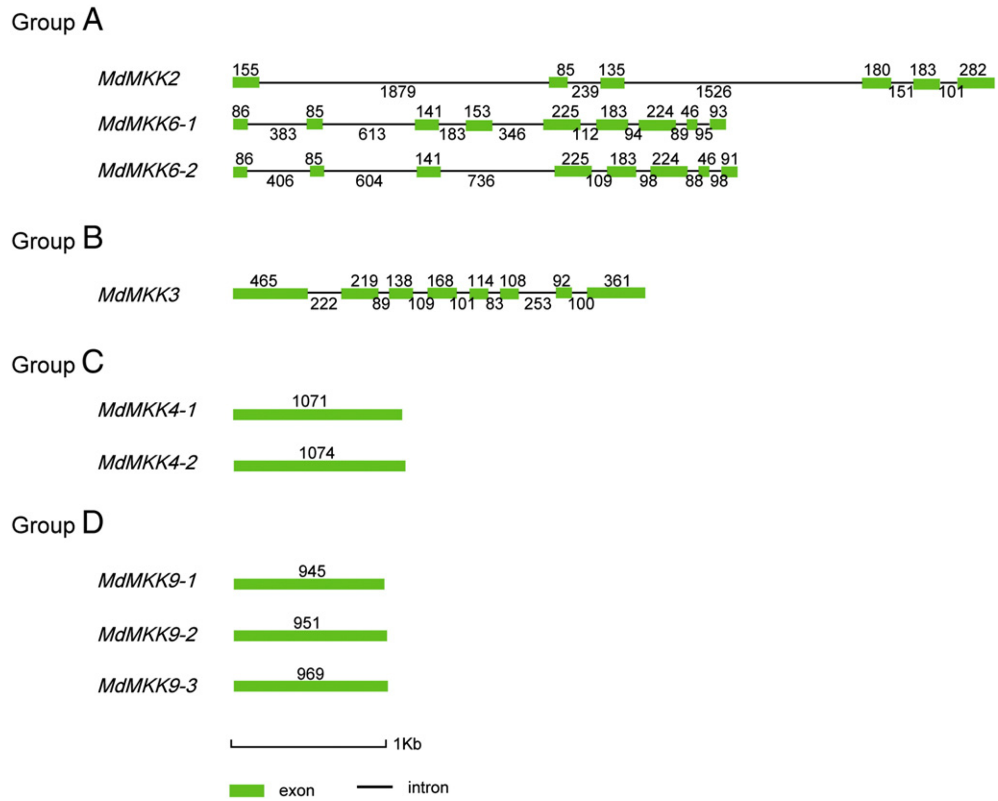

Fig. 4. The exon/intron structure of Apple *MAPKK* genes. Introns and exons are represented by black lines and green boxes respectively. The length in base pairs of each intron and exon is also indicated. Numbers correspond to the length of the intron and exon.

is well conserved not only between close paralogs, but also between the *MdMAPKs* that apparently diverged following earlier duplication events.

The *MdMAPKK* genes display two strikingly different structural patterns (Fig. 4), which is very similar to the *Arabidopsis* and poplar *MAPKK* genes. Members of group C and D *MAPKKs* have a completely intron less configuration in apple, *Arabidopsis* and poplar, whereas the group A and B *MAPKKs* possess numerous exon and intron junctions. In group A, *MdMKK2* possesses six exons, which are the same to *AtMKK1*; *MdMKK6-1*, two *AtMKKs* (*AtMKK2* and *AtMKK6*) and two PtMKKs (*PtMKK2-2* and *PtMKK6*) possess eight exons; and *MdMKK6-2* and *PtMKK2-1* possess nine exons. *MdMKK6-1* shows quite strong exon length conservation with *MdMKK6-2*. The only one member of group B, *MdMAPKK* (*MdMKK3*), possesses eight exons (Fig. 4), which is the same with *AtMKK3* in *Arabidopsis*. Overall, most closely related members in same subfamilies share similar exon/intron structure in terms of intron number and exon length.

3.4. Chromosomal location of MdMAPKs and MdMAPKKs on apple genomes

Chromosomal location analyses showed that 26 *MdMAPK* genes presented on 12 among total 17 chromosomes, which dispersed throughout their respective genomes (Fig. 5). Only one *MdMAPK* gene was found on each of chromosomes 9, 12, 14, 15, 16 and 17, while 2 genes on chromosomes 8 and 13, as well as 3 or more on chromosomes 1, 2, 3 and 11. Interestingly, four pairs of *MdMAPKs* (*MdMPK4-1/4-2*, *MdMPK3-1/3-3*, *MdMPK1-1/1-2* and *MdMPK7-1/7-2*) and a cluster (*MdMPK19-1*, *MdMPK19-2* and *MdMPK19-4*) were tightly co-located in apple genome. The other four sister pairs (*MdMPK6-1/MdMPK6-2*, *MdMPK13-1/MdMPK13-2*, *MdMPK9/MdMPK11* and *MdMPK20-1/MdMPK20-2*) were located on different chromosomes. Therefore, it is suggested that segmental duplication and transposition events all played roles in the evolution of *MAPK* gene family in apple.

MdMAPKK genes also display a scattered genomic distribution (Fig. 6) across six of the 17 apple chromosomes. Only chromosomes 2 and 9 contain two *MdMAPKK* genes, others contain only one gene. Three sister pairs (*MdMKK4-1/MdMKK4-2*, *MdMKK6-1/MdMKK6-2* and *MdMKK9-1/MdMKK9-3*) were located on different chromosomes. Therefore, segmental duplication events have played roles in the evolution of *MAPKK* gene family in apple.

3.5. Expression pattern of MdMAPKs and MdMAPKKs from microarray

The expression pattern of genes can provide important information for understanding their functions. Upon the microarray data of GEO (GSE24523), transcriptional variation of apple genes can be known from 4-week before ripening to ripening. Using BLAST search against the apple unigene database from the GDR database, 13 unigenes representing 20 *MdMAPKs* were identified (Fig. 6A). Based on hierarchical clustering, the expression patterns of the *MdMAPK* genes were divided into four groups: group I, group II, group III and group IV (Fig. 6A). Group I contains two group A *MdMAPK* genes, group II contains group B, C and D *MdMAPK* genes, group III contains group A and C *MdMAPK* genes, and the group IV contains group B and D *MdMAPK* genes. Group I genes exhibited high expression level at 4 weeks before ripening in CP. In contrast, genes from group II displayed high expression value at 0 week (ripening). Only five unigenes representing 7 *MdMAPKKs* were identified (Fig. 6B). Three pair genes (*MdMKK4-1/MdMKK4-2*, *MdMKK6-1/MdMKK6-2* and *MdMKK9-1/MdMKK9-3*) show the similar expression patterns.

Additionally, using BLAST search against the probe sequences (GPL3715), the expression patterns of 21 *MdMAPK* and 4 *MdMAPKK*-containing probers from microarray during rootstock–scion interactions process were found. Based on hierarchical clustering, the expression patterns of the *MdMAPK* genes were divided into four groups, named as group 1, group 2, group 3 and group 4 (Fig. 6C) and contain

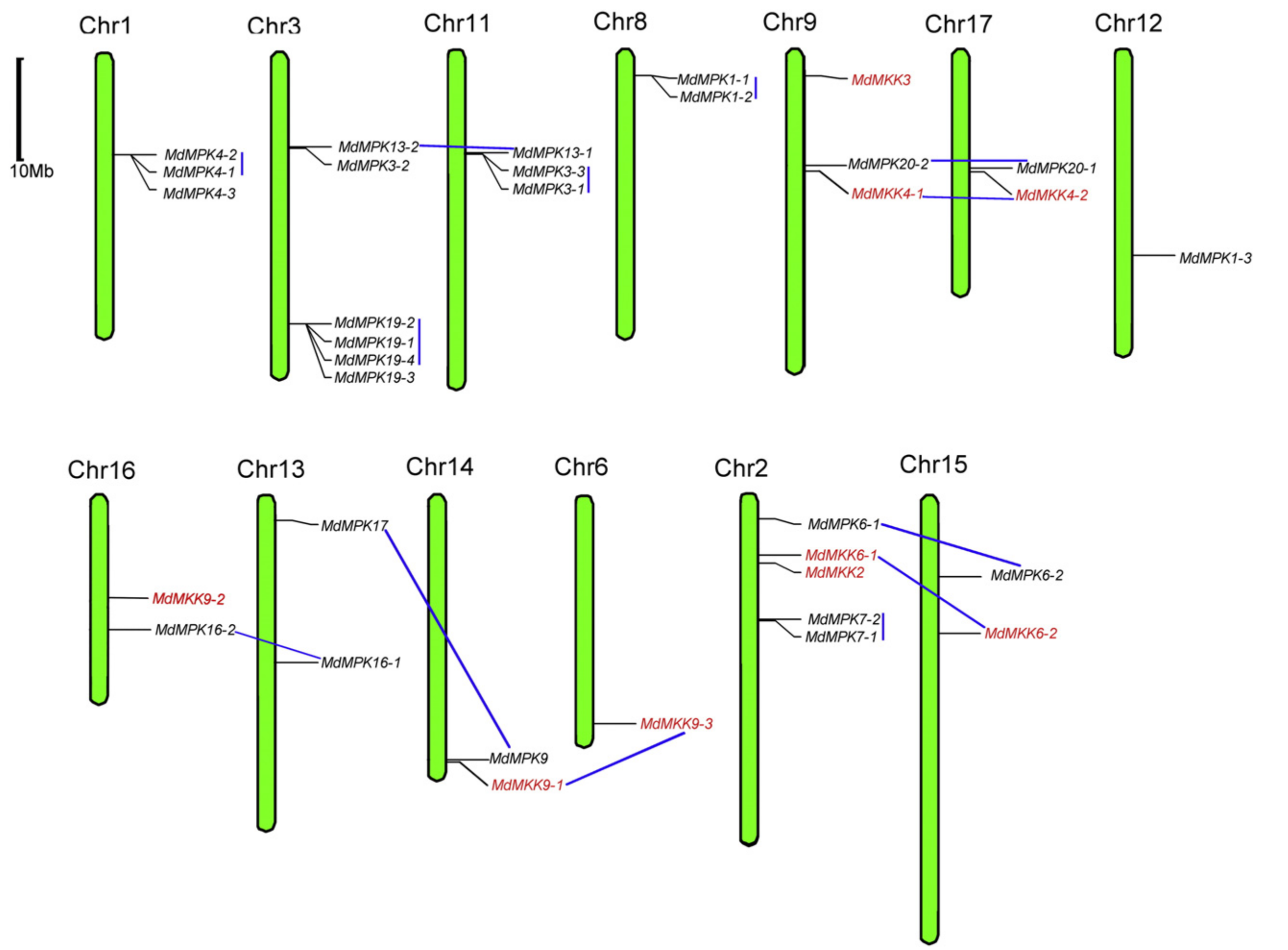

Fig. 5. Positions of *MAPK* and *MAPKK* gene family members on the apple chromosomes. The chromosome number is indicated at the top of each chromosome representation. Sister paralogous pairs are indicated by blue line. Scale represents a 10 Mb chromosomal distance.

8, 5, 2 and 6 MdMAPKs, respectively. As shown in Fig. 6D, only four probers representing 4 *MdMAPKKs* were identified and one pair of genes (*MdMKK4-1*/*MdMKK4-2*) shows the similar expression patterns. Overall, *MdMAPKs* and *MdMAPKKs* display the different expression patterns indicating that they may play different roles during fruit development and rootstock–scion interactions process.

3.6. Expression pattern of MdMAPKs and MdMAPKKs in various tissues as determined by semi-quantitative RT-PCR analyses

To study the function of the *MdMAPKs* and *MdMAPKKs* in different plant tissues (roots, stems, leaves, flowers and fruits), semi-quantitative RT-PCR was performed. As shown in Fig. 7, among the 26 *MdMAPKs*, 8 genes were expressed in all tissues tested with various expression levels, 17 showed different tissue-specific expression profiles, whereas *MdMPK17* showed too low transcript abundances to be observed in the above tissues suggesting it might be pseudogenes, or might be expressed at specific developmental stages, or under special conditions. It was noteworthy that *MdMPK13-2* and *MdMPK20-2* were exclusively expressed in roots, while *MdMPK1-1* was preferentially expressed in leaves although very low transcript abundances were detected in vegetative organs as well. All 9 *MdMKKs* were expressed in all tissues tested with various expression levels, although 3 *MdMKKs* (*MdMAKK9-1*, *MdMAKK9-2* and *MdMAKK9-3*) showed very low transcript abundances to be observed in the five tissues. Overall, these results indicated that *MdMAPKs* and *MdMAPKKs* played multiple roles in the development of apple.

4. Discussion

MAPK signal transduction modules play important roles in diverse processes, including developmental programs, defense and abiotic stress responses (Burnett et al., 2000; Cardinale et al., 2002; Colcombet and Hirt, 2008; Ichimura et al., 2000; Kiegerl et al., 2000; Kumar et al., 2008; Zhang et al., 2012a; Mockaitis and Howell, 2000; Munnik and Meijer, 2001; Raina et al., 2012; Ren et al., 2002; Teige et al., 2004; Yuasa et al., 2001; Z. Zhang et al., 2012). Besides, the presence of *MAPK* and *MAPKK* genes in land plants ranging from mosses to eudicots also makes them interesting candidates for the evolution of plant development (Colcombet and Hirt, 2008; Zhang et al., 2012a; Meldau et al., 2012; Mockaitis and Howell, 2000; Raina et al., 2012; Sasabe et al., 2006; Zhang et al., 2012b). To date, a conserved number of *MAPK* and *MAPKK* have so far been identified and functionally characterized in both model and crop plants such as *Arabidopsis*, rice, poplar and tomato (Asai et al., 2002; Chen et al., 2012; Colcombet and Hirt, 2008; Kong et al., 2012; Nicole et al., 2006; Q. Kong et al., 2012; Rao et al., 2010; Zhang et al., 2013). However, no data set of *MAPK* and *MAPKK* is available for apple.

In this study, we first identified a total of 26 *MAPKs* and 9 *MAPKKs* containing full ORFs in apple through genome-wide analysis, respectively. *MAPK* and *MAPKK* gene families are conserved in the plants, such as 20 *MAPKs* and 10 *MAPKKs* in *Arabidopsis*, 15 *MAPKs* and 8 *MAPKKs* in rice, as well as 21 *MAPKs* and 11 *MAPKKs* in poplar. The *MAPK* gene family in apple is by far the largest one compared to the estimates for other plant species, indicating that *MAPK* gene family in

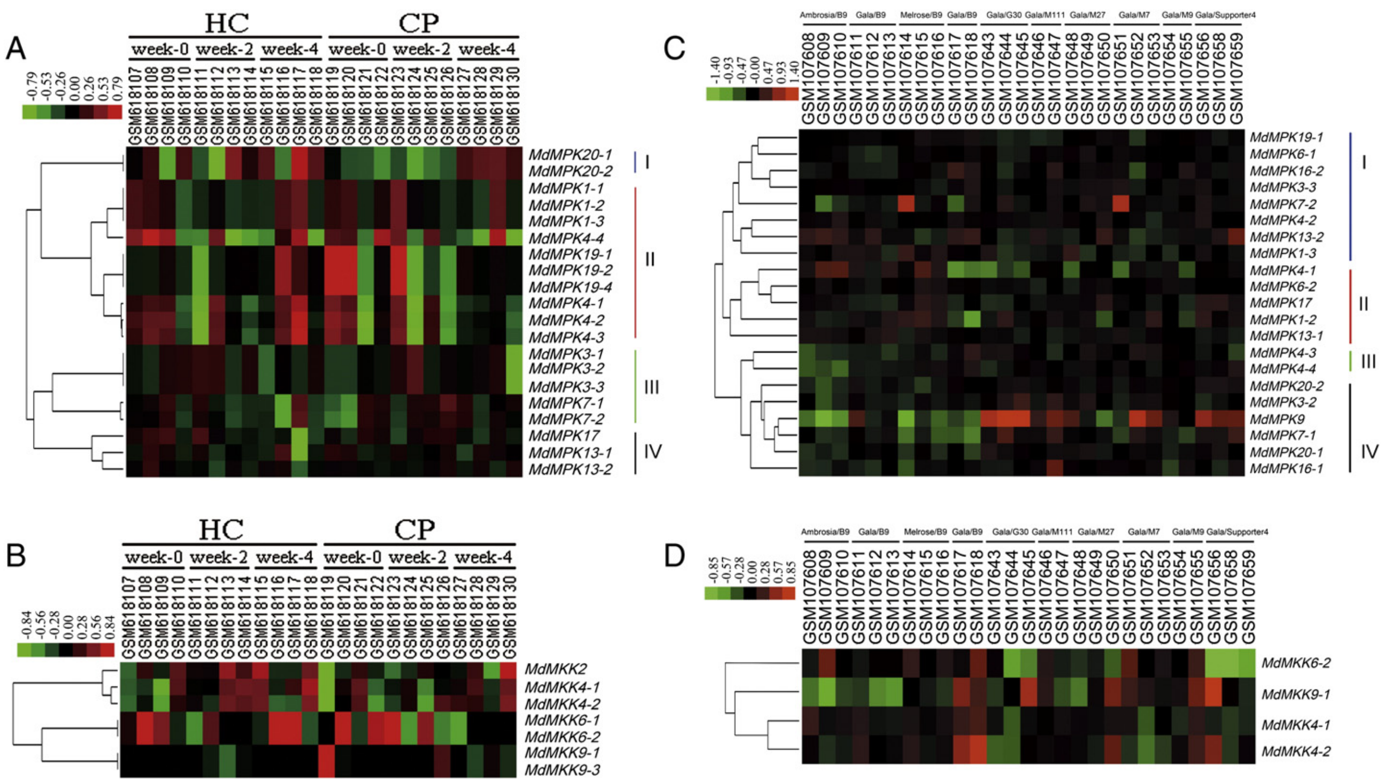

Fig. 6. The expression patterns of the *MdMAPKs* and *MdMAPKKs* from microarray. A, The expression patterns of the *MdMAPKs* from microarray during fruit development in apple. B, The expression patterns of the *MdMAPKKs* from microarray during fruit development in apple. C, The expression patterns of the *MdMAPKs* from microarray during rootstock–scion interactions process. D, The expression patterns of the *MdMAPKKs* from microarray during rootstock–scion interactions process. The color scale representing the relative signal values is shown above (green refers to low expression; black refers to medium expression and red refers to high expression). HC, Honeycrisp; CP, Cripps Pink. Week-4, 4 weeks before ripening; Week-2, 2 weeks before ripening; Week-0, ripening. 24 samples are shown as GSM618107–GSM618130 (GEO accession numbers). 27 samples (GSM107608–GSM107659) are ten types for rootstock–scion interactions (Ambrosia/B9, Gala/B9, Melrose/B9, Gala/B9, Gala/G30, Gala/M111, Gala/M27, Gala/M7, Gala/M9 and Gala/Supporter).

apple has expanded. It can be speculated that the presence of more *MAPK* genes in apple genome may reflect the great needs for these genes to be involved in the complicated transcriptional regulations in this woody perennial species. Meanwhile, we also use the web resources to predict the proteins property, such as the isoelectric point and molecular weight.

To examine the phylogenetic relationships among the MAPK proteins in apple and *Arabidopsis*, an unrooted tree was constructed from alignments of the MAPK protein sequences. Given that the highly divergent sequences were responsible for the regulation diversities of MAPK proteins, we constructed the phylogenetic tree based on the full length of the MAPK proteins. Four subgroups were clustered by the phylogenetic tree, and named as groups A to D just as the previous studies (MAPK Group, 2002). Using the same method, the *MdMAPKKs* were divided into four subfamilies (A–D). As shown in Figs. 1 and 2, MAPK or MAPKK genes with same functions showed a tendency to cluster into one subgroup, which provided an excellent reference to explore the functions of the *MdMAPKs*. Additionally, most closely related members in same subfamilies share similar exon/intron structure in terms of intron number and exon length among *Arabidopsis*, rice, tomato, poplar and apple.

Gene duplications have an important role not only in the genomic rearrangement and expansion but also in diversification of gene function. It has been reported that a recent GWD event happened in apple 60–65 million years ago, which resulted in the expansion of several gene classes (Velasco et al., 2010). According to the past report, gene duplication caused expansion of some gene families (Ring, LBD, HSP and DREB) (Giorno et al., 2012; Li et al., 2011; Liang et al., 2012; Zhao et al., 2012). In this study, chromosomal location analysis showed that *MAPK* and *MAPKK* genes of apple were dispersed throughout 13 out of 17 chromosomes with different densities. Four sister pair *MdMAPKs* and a cluster (*MdMPK19-1*, *MdMPK19-2* and *MdMPK19-4*) were tightly co-located in apple genome, while five sister pair *MdMAPKs* and three sister pair *MdMAPKKs* were located on different chromosomes. As well as, multiple pairs linked each of at least 15 potential of chromosomal segmental duplications, such as the large sections of Chr 3 and 11, 4 and 12. Consistent with this, there was a clear paralogous pattern of gene family divergence by gene duplication for the apple. Evolutionary divergence analysis suggested that the whole genome and chromosomal segment duplications and transposition events might contribute to the expansion of *MdMAPKs* and *MdMAPKKs*.

Microarray and expressed sequence tag (EST) data revealed the expression patterns of 20 *MdMAPK* genes and 5 *MdMAPKK* genes during fruit development, as well as 21 *MdMAPK* and 4 *MdMAPKK* genes from microarray during rootstock–scion interactions process. According to the EST analysis, *MdMAPK* genes and *MdMAPKK* may play different roles during fruit development and rootstock–scion interactions process. Since gene expression patterns can provide important clues for gene function, we examined the expression of all the *MAPK* and *MAPKK* genes in root, stem, leaf, flower and fruit tissues using semi-quantitative RT-PCR. With the exception of one *MAPK* gene showing very low or no transcript abundances in the above tissues, all the other genes displayed diverse expression patterns, representing distinct roles of individual *MdMAPKs* and Md*MAPKKs*. Interestingly, two *MdMAPK* genes (*MdMPK13-2* and *MdMPK20-2*) were found to be mainly expressed in roots, indicating that these genes may be involved in the regulation of root development. In future, further study will be needed to determine the functions of the *MdMAPKs* and Md*MAPKKs* genes in apple. Taken together, these results indicated that *MdMAPKs* and Md*MAPKKs* played a major role in the development of apple.

Overall, in this study we presented a complete analysis of *MAPK* and *MAPKK* family in apple, with a special emphasis on fruit development

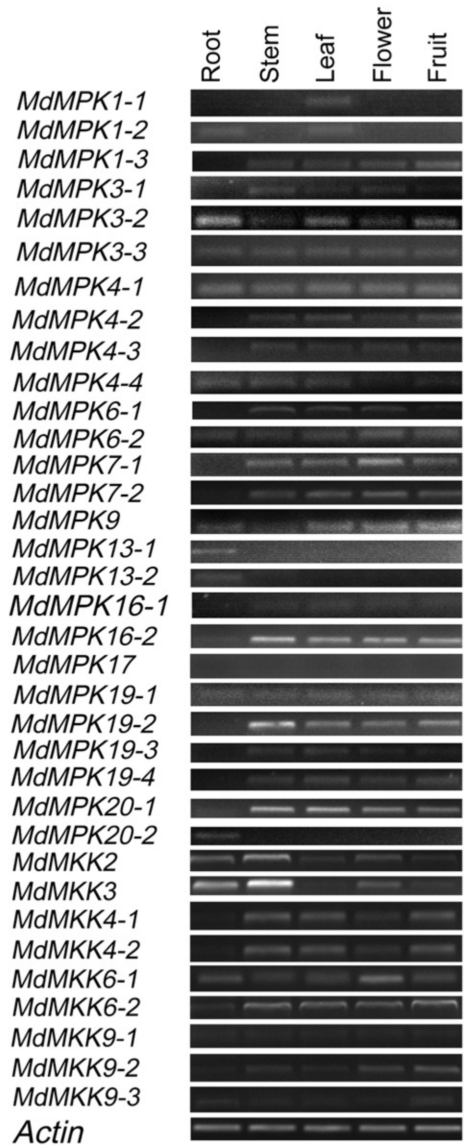

Fig. 7. Tissue-specific expression profiles for the *MdMAPK* and *MdMAPKK* genes. Expression levels of *MdLBD* genes were examined by semi-qRT-PCR in apple roots, stems, leaves, flowers and fruits. The *MdACTIN* was performed as an internal control.

and rootstock–scion interactions process. Our results presented here would be helpful in laying the foundation for functional characterization of *MAPK* and *MAPKK* gene family and further gaining an understanding of the structure–function relationship between these family members. Additionally, our study provides comprehensive information and novel insights into the evolution and divergence of the *MAPK* and *MAPKK* genes in plants. Moreover, these studies may potentially aid in the understanding of the molecular basis of many agronomically important traits of apple, such as fruit development and other physiological process.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.gene.2013.07.107.

Conflict of interest

There is no conflict of interest.

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